

Haripriya Varanasi

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SUMMARY

Data Analyst / Business Analyst with a Master of Science in Data Analytics Engineering and hands-on experience across Python, SQL, DuckDB, Power BI, Tableau, AWS, Snowflake, and machine learning. Track record of turning open-ended business problems into decision-support tools, dashboards, and predictive models across financial, operational, environmental, and public-sector domains. Authorized to work in the U.S. under F-1 OPT; future H-1B sponsorship may be required.

EXPERIENCE

AI Data Solutions Analyst

Blue Horse Digital

Jul 2026 – Present

Remote

- Conducted an independent financial and operational analysis of USPS using its OIG report, identifying a \$11.7B FY23 expense variance against a \$5.6B revenue variance using Python, pandas, and Matplotlib, and delivering a KPI framework and forecasting recommendations to leadership
- Contributed research and strategic framing to a team AI-automation feasibility proposal for USPS OIG, validating public data sources for a no-access prototype path and scoping human-in-the-loop tool boundaries

Graduate Research Assistant

George Mason University

Aug 2024 – May 2026

Fairfax, VA

- Reviewed and validated 40+ graduate-level analytics projects for methodological rigor (data leakage, class imbalance, model validation, evaluation metrics), holding weekly office hours and live debugging sessions
- Authored a shared data-quality rubric and structured feedback framework that standardized project review consistency across the course

Data Analyst / Scrum Master

Gulfstream Foundation (GMU Capstone)

Jan 2026 – May 2026

Fairfax, VA

- Built the Risk Rankings, Live Conditions, and Chatbot pages of a Streamlit flood-risk dashboard covering all 78 Puerto Rico municipios, integrating 10 federal data sources via a DuckDB-backed pipeline automated with GitHub Actions
- Served as Scrum Master for a 6-person team, running sprint planning and stakeholder check-ins; presented the finished platform at STAR-TIDES 2026

Student Services Administrative Intern

Andhra University

Aug 2023 – May 2024

Visakhapatnam, India

- Maintained academic-performance records using Excel PivotTables and structured reporting; mentored peers in Python, Jupyter, and RStudio for coursework and technical projects
- Coordinated 15+ academic, wellness, and onboarding events, standardizing templates and documentation practices across administrative records

AWS Cloud Intern

BrainOVision Solutions India Pvt. Ltd.

Jul 2022 – Jul 2023

Hyderabad, India

- Completed hands-on training across AWS EC2, S3, IAM, RDS, and Lambda, building small ETL workflows connecting cloud storage, compute, and database components

PROJECTS

CFPB Consumer Complaint Decision-Support Platform | *Python, DuckDB, Streamlit, scikit-learn, Ollama*

- Built a local-first platform querying 16M+ CFPB records, including chronologically-validated Random Forest models (ROC-AUC 0.69), semantic search, and a local-LLM natural-language-to-SQL feature with 12 automated safety tests

US Air Quality Analytics Pipeline | *Python, pandas, AWS S3, Tableau Public*

- Built an end-to-end pipeline processing 2024 EPA Daily AQI-by-county data; identified California as home to 6 of the 10 most polluted counties and validated a legitimate 1,513 AQI outlier in Essex County, MA

Obesity Level Prediction | *Python, scikit-learn, XGBoost, AIT 664 course project*

- Compared Logistic Regression, Random Forest, and XGBoost on WHO obesity data; achieved 0.990 macro-F1 with Random Forest and showed BMI/lifestyle features were far stronger predictors than technology use alone (0.18 macro-F1)

SmartPave: Pavement Deterioration Modeling | *PySpark, Databricks, AWS, XGBoost*

- Built a distributed regression pipeline on 200K+ FHWA pavement records; XGBoost reached R^2 of 0.64–0.87 across three research questions on climate, traffic, and geographic drivers of cracking

Life Expectancy 360: A Holistic Data Perspective | *R, Multiple Linear Regression, GBUS 738 course project*

- Led data preparation and EDA for a 4-person team; full regression model explained 97.7% of variance in life expectancy across GDP, mortality, and education factors from WHO data (2,864 records)

Modern Analytics Engineering Warehouse | *Snowflake, dbt Core, GitHub Actions*

- Built a governed dbt/Snowflake warehouse from 1M+ retail records with automated testing and CI, replacing 20+ ad hoc queries with 6 reusable marts and cutting modeled transformation time 60–75%

TECHNICAL SKILLS

Languages/Query: Python, SQL, DuckDB SQL, R, PySpark

Data & ML: pandas, NumPy, scikit-learn, XGBoost, Random Forest, Logistic Regression, PCA, K-Means, regression, classification, feature engineering, cross-validation

BI & Visualization: Power BI, DAX, Tableau, Streamlit, Plotly, Matplotlib, Excel

Cloud & Data Platforms: AWS (S3, EC2, IAM, Lambda, RDS), Snowflake, dbt, Databricks, GitHub Actions

AI/NLP: Ollama, sentence-transformers, semantic search, natural-language-to-SQL, prompt engineering

EDUCATION

George Mason University

Master of Science in Data Analytics Engineering; GPA: 3.73/4.0

Fairfax, VA

Aug 2024 – May 2026

PROFESSIONAL DEVELOPMENT

Women Engineers Program – TalentSprint, supported by Google